
Technical Specifications

22 August 2014

FIX 4.2 Protocol Specification Guide

Version 3.2



London
Stock Exchange Group

Table of Contents

1.0	Introduction	5
1.1	Purpose	5
1.2	Readership	5
1.3	Revision History	5

2.0	Message Header and Trailer	6
2.1	Standard Message Header	6
2.2	Standard Message Trailer	7

3.0	Administrative Messages	7
3.1	A: Logon	7
3.2	O: Heartbeat	7
3.3	1: Test Request	8
3.4	2: Resend Request	8
3.5	3: FIX Protocol Error / Reject	8
3.6	4: Sequence Reset	9
3.7	5: Logout	9

4.0	Application Messages	10
4.1	AE: Trade Capture Report	10
4.2	AF: Order Mass Status Request	12
4.3	AR: Trade Capture Report Acknowledgment	13
4.4	D: New Order - Single	16
4.5	G: Order Cancel/Replace Request (Order Modification Request)	18
4.6	F: Order Cancel Request	20
4.7	R: Quote Request	21
4.8	b: Quote Acknowledgement	22
4.9	c: Security Definition Request	22
4.10	d: Security Definition	24
4.11	j: Business Reject	27
4.12	s: New Order Cross	27
4.13	8: Execution Report	29
4.14	9: Order Cancel Reject	34

5.0	Field Definitions	35
6.0	Error Codes	50
7.0	Message Flow	53
7.1	Connection Management	53
	Establishing a FIX Session	53
	Initializing a FIX Connection	54
	Terminating a FIX Connection	54
	Sending a Heartbeat	54
	Heartbeat Management	54
	Sending a Resend Request	54
	Sending a Sequence Reset	55
	Receiving a Session Level Reject	55
	Using a Single FIX Session to Send Orders for Multiple Traders/Users	56
7.2	Order Processing	56
	Order is rejected	57
	Order is accepted and fully executed	58
	Minimum Quantity Order is not executed when entered	58
	New Order is sent with Possible Resend	58
	Order is resent with Possible Resend	59
	Stop and If Touched Order	59
	Stop Order is triggered	60
	Immediate Order Completely Filled upon entry	60
	Immediate Order partially Executed in Continuous Trading	60
	Immediate is not Executed in Continuous Trading	61
7.3	Time Validity	62
	While Connected orders cancelled on disconnection of a participant	62
	While Connected orders cancelled on EOD Mini batch	62
7.4	Cross/Committed Functionality	63
	Committed orders traded	63
	Committed order cancelled by participant before it trades	64
	Pending Committed order cancelled during EOD	64

Committed order not accepted by the trading engine	65
Entering an Accepted Cross Order	65
Cross Order Rejected	65
7.5 Modification Processing	66
Modification is accepted	66
Modification is rejected	67
Cancellation is accepted	67
Cancellation is rejected	67
7.6 Strategy Messages	68
User defined Strategy (FLEXCO) Creation Accepted	68
User defined Strategy (FLEXCO) Creation Error	68
7.7 Quote Processing	69
Request for Quote	69
7.8 Unsolicited Services	69
Elimination of an Order	69
Global Cancellation of all Orders for a Member Initiated by the Exchange	70
Cancellation of a Trade by the Exchange	70
Cancellation of a Trade with Impact on the Last Price Made by the Exchange	71
7.9 Queries	71
Order Mass Status	71
Security Definition	71

1.0 Introduction

1.1 Purpose

The purpose of this publication is to provide participants with the knowledge and technical details necessary for accessing and using the LSEG's derivatives trading system.

This FIX specification provides essential information for participants and independent software vendors in the functional design of their application in order to interface with the Exchange's derivatives platform using the Financial Information eXchange (FIX) Protocol. This document defines the subset of the Financial Information eXchange (FIX) messages that are supported by the Exchange on its FIX trading interface.

The derivatives platform utilises FIX 4.2 with a few exceptions as specified in this document. The FIX interface does not provide functions related to Market Making. Participants who intend to be Market Makers must use the native SOLA Access Information Language (SAIL) protocol.

This document is designed to supplement the FIX protocol documentation that can be found at www.fixprotocol.org rather than be a complete and self-sufficient reference.

Note: The only FIX messages and fields accepted by the Exchange are the ones described in this document. Unsupported fields are rejected.

1.2 Readership

The target audience for this publication is the business or Information Technology level of an organisation interested in the functional design of the LSEG's derivatives platform.

1.3 Revision History

This document has been through the following iterations:

Issue	Date	Description
1.0	21 December 2010	Publication of initial version
1.1	04 April 2011	Update to initial version
1.2	16 May 2011	Accuracy adjusted to 4 decimal places from 6 decimal places
2.0	July 2012	Message modifications and new error codes introduced with the updated version for Sola 5
2.1	March 2014	Added the Message Flow section. Corrected "Order Type" specification with all possible markers.
3.0	October 2014	Added Sola 7 Message Impacts
3.1	November 2014	Added Self Execution Prevention (SEP) impacts
3.2	December 2014	Reviewed version

2.0 Message Header and Trailer

For additional descriptive or definitive information on Tag Numbers and Field Names, refer to Field Definitions.

2.1 Standard Message Header

Tag	Field Name	Req	Comments
8	BeginString	Y	FIX.4.2 Must be first field in message.
9	BodyLength	Y	Must be second field in message.
35	MsgType	Y	Must be third field in message.
49	SenderCompID	Y	Assigned value used to identify the sender in a FIX session
50	SenderSubID	N	Assigned value used to identify specific message originator (desk, trader, etc.).
56	TargetCompID	Y	Assigned value used to identify the receiver in a FIX session.
57	TargetSubID	N	Assigned value used to identify specific individual or unit intended to receive message.
34	MsgSeqNum	Y	Message sequence number.
52	SendingTime	Y	Time of message transmission.
43	PossDupFlag	N	Always required for retransmitted messages, whether prompted by the sending system or as the result of a resend request.
122	OrigSendingTime	N	Required for message resends. If data is not available, set to same value as SendingTime.
97	PossResend	N	Required when message may be duplicate of another message sent under a different sequence number.

2.2 Standard Message Trailer

Tag	Field Name	Req	Comments
10	Checksum	Y	Always unencrypted and last field in message.

3.0 Administrative Messages

3.1 A: Logon

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=A]
98	EncryptMethod	Y	Must be set to 0.
108	HeartBtInt	Y	0 (zero) means that no HeartBeat message will be sent. The value provided must not be less than 30 (unless using 0).
141	ResetSeqNumFlag	N	Indicates both sides of a FIX session should reset sequence numbers.
383	MaxMessageSize	N	Can be used to specify the maximum number of bytes supported for messages received.
	Standard Trailer	Y	

3.2 0: Heartbeat

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=0]

Tag	Field Name	Req	Comments
112	TestReqID	N	Required when the Heartbeat is the result of a Test Request message. Used only in Heartbeat message from server to client.
	Standard Trailer	Y	

3.3 1: Test Request

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=1]
112	TestReqID	Y	Identifier included in Test Request message to be returned in resulting Heartbeat
	Standard Trailer	Y	

3.4 2: Resend Request

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=2]
7	BeginSeqNo	Y	Message sequence number of first message in range to be resent
16	EndSeqNo	Y	Message sequence number of last message in range to be resent.
	Standard Trailer	Y	

3.5 3: FIX Protocol Error / Reject

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=3]
45	RefSeqNum	N	MsgSeqNum of rejected message.

Tag	Field Name	Req	Comments
371	RefTagID	N	The Tag number of the FIX field being referenced.
372	RefMsgType	N	The MsgType of the FIX message being referenced.
373	SessionRejectReason	N	Code to identify reason for a session-level. Reject message.
58	Text	N	Where possible, message to explain reason for rejection.
	Standard Trailer	Y	

3.6 4: Sequence Reset

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=4]
123	GapFillFlag	N	Indicates that the Sequence Reset message is replacing administrative or application messages which will not be resent.
36	NewSeqNo	Y	Cannot be lower than the last [MsgSeqNum 34].
	Standard Trailer	Y	

3.7 5: Logout

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=5]
58	Text	N	Free format text string.
	Standard Trailer	Y	

4.0 Application Messages

4.1 AE: Trade Capture Report

This message is used for pre-arranged trade(s) with multiple counterparties. This message has been taken from FIX 5.0 and adapted to SOLA platform.

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=AE]
60	TransactTime	N	Time at which this order request was initiated/released by the trader or trading system
16453	ProposalType	Y	B: Bundle Order T: Inter Dealer Broker Order
571	TradeReportID	N	Proposal ID For messages coming from Exchange
1117	RootPartyID	N	Required when ProposalType 16453 = T: Inter Dealer Broker Order
150	ExecType	N	Present in messages coming from Exchange Proposal Status: 0: New (Waiting for approval) 4: Cancelled (Refused) F: Trade (Approved)
1041	FirmTradeID	Y	The ID assigned to a trade by the Firm to track a trade within the Firm system.
555	NoLegs	Y	Number of legs Up to 4 for messages coming from Participant
N Times (NoLegs)			
600	LegSymbol	Y	Mandatory as the first Tag present for the Leg. Indicates the instrument root Symbol

Tag	Field Name	Req	Comments
613	LegOptAttribute	N	Corporate Action Marker
609	LegSecurityType	Y	Indicates if Option (OPT) or a Future (FUT)
610	LegMaturityMonthYear	Y	Required for Options (LegSecurityType 609=OPT) and Futures(LegSecurityType 609=FUT)
16454	LegMaturityDay	Y	Can be used in conjunction with MaturityMonth-Year <200> to specify a particular maturity date.
1358	LegPutOrCall	N	Required for Options (LegSecurityType 609=OPT)
612	LegStrikePrice	N	Required for Options (LegSecurityType 609=OPT)
1194	OptionStyle	N	Required for Options (LegSecurityType 609=OPT) 0: European 1: American
687	LegQty	Y	Order quantity
566	LegPrice	Y	Order Price
624	Leg Side	Y	1: Buy 2: Sell
37	Order ID	N	Present in messages coming from the Exchange
448	PartyID	Y	Executing Firm ID
577	ClearingInstruction	N	Clearing instruction. Free text field (up to 12 chars). Required for messages coming from Participant when Proposal Type 16453 = B: Bundle

Tag	Field Name	Req	Comments
581	CaptRptAccountType	N	Type of account Required for messages coming from Participant when Proposal Type 16453 = B: Bundle
77	OpenClose	N	Open/Close Required for messages coming from Participant when Proposal Type 16453 = B: Bundle
8001	AccountProfile	N	Type of trader. H: Hedger S: Speculator Required for messages coming from Participant when Proposal Type 16453 = B: Bundle
9433	TextExecutionReport	N	Owner Data for messages coming from Participant when 16453 = B Indicates the Refusal Reason when ExecType 60 = 4: Cancelled
852	PublishTrdIndicator	N	Transparency for the Flexible trade. Y: Published N: Unpublished
39	OrdStatus	Y	0: New (Waiting for approval) 2: Fill (In Order Book) Only for messages coming from Exchange
11	ClOrdID	Y	Must be unique for messages coming from Participant. Not to be specified when ProposalType 16453 = T: Inter Dealer Broker Order.
	Standard Trailer	Y	

4.2 AF: Order Mass Status Request

Execution Report [MsgType 35=8] with [ExecTransType 20=3] (Status) are returned for each active order belonging to the participant. If no active order belongs to the participant, no response is returned.

Tag	Field Name	Req	Comments
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Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=AF]. The fields "SenderSubId" and "TargetSubId" of the standard header should not be sent, otherwise Sola will reject the message.
584	MassStatusReqID	N	Value assigned by issuer of Mass Status Request to uniquely identify the request.
585	MassStatusReqType	N	= 7 (Status for all orders for the participant)
	Standard Trailer	Y	

4.3 AR: Trade Capture Report Acknowledgment

This message is used for accepting or refusing pre-arranged trade(s) with multiple counterparties. This message has been taken from FIX 5.0 and adapted to SOLA platform.

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=AR]
60	TransactTime	N	Time at which this order request was initiated/released by the trader or trading system
16453	ProposalType	Y	B: Bundle Order T: Inter Dealer Broker Order
571	TradeReportID	Y	Proposal ID
856	TradeReportType	Y	2: Accept 3: Decline

Tag	Field Name	Req	Comments
1041	FirmTradeID	Y	The ID assigned to a trade by the Firm to track a trade within the Firm system.
555	NoLegs	N	Number of legs Required for messages coming from Participant when TradeReportType 856 = 2: Accept
N Times (NoLegs) - The following fields are required for messages coming from Participant when TradeReportType 856 = 2: Accept			
600	LegSymbol	Y	Mandatory as the first Tag present for the Leg. Indicates the instrument root Symbol
613	LegOptAttribute	N	Corporate Action Marker
609	LegSecurityType	Y	Indicates if Option (OPT) or a Future (FUT)
610	LegMaturityMonthYear	Y	Required for Options (LegSecurityType 609=OPT) and Futures(LegSecurityType 609=FUT)
16454	LegMaturityDay	Y	Required for Options (LegSecurityType 609=OPT) and Futures(LegSecurityType 609=FUT) Can be used in conjunction with MaturityMonth-Year <200> to specify a particular maturity date.
1358	LegPutOrCall	N	Required for Options (LegSecurityType 609=OPT)
612	LegStrikePrice	N	Required for Options (LegSecurityType 609=OPT)
1194	OptionStyle	N	Required for Options (LegSecurityType 609=OPT) 0: European 1: American
687	LegQty	Y	Order quantity

Tag	Field Name	Req	Comments
566	LegPrice	N	Order Price. Required for messages coming from Participant when TradeReportType 856 = 2: Accept.
624	LegSide	Y	1: Buy 2: Sell
37	Order ID	Y	Required for all messages coming from Participant
448	PartyID	N	Executing Firm ID. Required for messages coming from Participant when TradeReportType 856 = 2: Accept
577	ClearingInstruction	N	Clearing instruction. Free text field (up to 12 chars). Required for messages coming from Participant when TradeReportType 856 = 2: Accept
581	CaptRptAccountType	N	Type of account Required for messages coming from Participant when TradeReportType 856 = 2: Accept
77	OpenClose	N	Open/Close Required for messages coming from Participant when TradeReportType 856 = 2: Accept
8001	AccountProfile	N	Type of trader. H: Hedger S: Speculator Required for messages coming from Participant when TradeReportType 856 = 2: Accept
9433	TextExecutionReport	N	Indicates the Refusal Reason. Required for messages coming from Participant when TradeReportType 856 = 3: Decline
852	PublishTrdIndicator	N	Transparency for the Flexible trade when TradeReportType 856 = 2: Accept Y: Published N: Unpublished
11	CIOrdID	N	Must be unique for messages coming from Participant. Required for messages coming from Participant when TradeReportType 856 = 2: Accept

Tag	Field Name	Req	Comments
	Standard Trailer	Y	

4.4 D: New Order - Single

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=D]
11	ClOrdID	Y	Unique identifier of the order as assigned by institution.
1	Account	N	Account mnemonic as agreed between broker and institution.
167	SecurityType	Y	Indicates if Option (OPT), a Future (FUT), or Strategy (STR).
55	Symbol	Y	Mandatory
201	PutOrCall	N	Mandatory for SecurityType 167=OPT: Option.
202	StrikePrice	N	Mandatory for SecurityType 167=OPT: Option.
200	MaturityMonthYear	N	Mandatory for SecurityType 167=OPT: Option and SecurityType 167=FUT: Future.
205	MaturityDay	N	Used in conjunction with MaturityMonthYear <200> to specify a particular maturity date.

Tag	Field Name	Req	Comments
206	OptAttribute	N	Indicates a corporate action
54	Side	Y	Buy or Sell.
60	TransactTime	N	Time at which this order request was initiated/released by the trader or trading system.
38	OrderQty	Y	Number of shares ordered.
40	OrdType	Y	Please refer to Field Definitions
44	Price	N	Required when [OrdType 40 = 2 or 4].
59	TimeInForce	N	Absence of this field indicates Day order.
432	ExpireDate	N	Conditionally required if [TimeInForce 59=6] (GTD)
58	Text	N	Free format text string.
77	Open/Close	Y	Please refer to Field Definitions
47	Rule80A	Y	Identify the type of account.
99	StopPx	N	Required when [OrdType 40 = 4]
110	MinQty	N	Minimum quantity of an order to be executed.

Tag	Field Name	Req	Comments
210	MaxShow	N	Maximum number of shares within an order to be shown to other customers
8001	AccountProfile	N	Indicate the type of trader
337	Contra Trader	N	Required if OrdType is C (Committed). This tag is the opposite firm when entering a committed order
5255	StopPxCondition	N	Required if Tag 99 (StopPx) is filled. This tag specifies the type of stop orders
	Standard Trailer	Y	

4.5 G: Order Cancel/Replace Request (Order Modification Request)

All fields can be modified except; [OrigClOrdID 41], [Symbol 55], [SecurityType 167], [PutOrCall 201], [StrikePrice 202], [MaturityMonthYear 200], [MaturityDay 205], [OptAttribute 206], and [Side 54].

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=G]
37	OrderID	N	Unique identifier of most recent order as assigned by the Exchange.
1	Account	N	Account mnemonic as agreed between broker and institution.
41	OrigClOrdID	Y	[ClOrdID 11] of the previous order (NOT the initial order of the day) when cancelling or replacing an order.
11	ClOrdID	Y	Unique identifier of replacement order as assigned by institution. Note that this identifier will be used in [ClOrdID 11] field of the Cancel Reject message if the replacement request is rejected.
167	SecurityType	Y	Indicates if Option (OPT), a Future (FUT), or Strategy (STR).

Tag	Field Name	Req	Comments
55	Symbol	Y	Must match original order. Mandatory
201	PutOrCall	N	Must match original order. Mandatory for Option.
202	StrikePrice	N	Must match original order. Mandatory for Option.
200	MaturityMonthYear	N	Must match original order. Mandatory for Option and Future.
205	MaturityDay	N	Used in conjunction with MaturityMonthYear <200> to specify a particular maturity date.
206	OptAttribute	N	Must match original order
54	Side	Y	Must match original side.
60	TransactTime	Y	Time this order request was initiated/released by the trader or trading system.
38	OrderQty	Y	Represent the remaining active quantity.
40	OrdType	Y	Please refer to Field definitions
44	Price	N	Required when [OrderType 40 = 2 or 4].
59	TimeInForce	N	Absence of this field indicates Day order.
99	StopPx	N	Required when [OrdType 40 = 4]
432	ExpireDate	N	Conditionally required if [TimeInForce 59=6] (GTD).
58	Text	N	Free format text string.
77	Open/Close	Y	Please refer to Field Definitions
47	Rule80A	Y	Identifies the type of account.
210	MaxShow	N	Maximum number of shares within an order to be shown to other customers
8001	AccountProfile	N	Indicate the type of trader
337	Contra Trader	N	Required if OrdType 40 = C (Committed) This tag is the opposite firm when entering a committed order
5255	StopPxCondition	N	Required if Tag 99 (StopPx) is filled. This tag specifies the type of stop orders

Tag	Field Name	Req	Comments
	Standard Trailer	Y	

4.6 F: Order Cancel Request

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=F]
41	OrigClOrdID	Y	[ClOrdID 11] of the previous order (NOT the initial order of the day) when cancelling or replacing an order.
37	OrderID	N	Unique identifier of most recent order as assigned by the Exchange.
11	ClOrdID	Y	Unique ID of cancel request as assigned by the institution.
38	OrderQty	N	Number of shares ordered.
167	SecurityType	Y	Indicates if Option (OPT), a Future (FUT), or Strategy (STR).
55	Symbol	Y	Must match original order.
201	PutOrCall	N	Must match original order. Mandatory for Option.
202	StrikePrice	N	Must match original order. Mandatory for Option.
200	MaturityMonthYear	N	Must match original order. Mandatory for Option and Future.
205	MaturityDay	N	Used in conjunction with MaturityMonthYear <200> to specify a particular maturity date.
206	OptAttribute	N	Indicates a corporate action

Tag	Field Name	Req	Comments
54	Side	Y	Buy or Sell
60	TransactTime	Y	Time this order request was initiated/released by the trader or trading system.
	Standard Trailer	Y	

4.7 R: Quote Request

This message is used to broadcast a request for a quote on a particular instrument. The request is broadcasted on the HSVF Market Data feed.

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=R]
131	QuoteReqID	N	Unique identifier for quote request.
146	NoRelatedSym	N	If provided, must be set to 1.
167	SecurityType	Y	Indicates if Option (OPT), a Future (FUT), or Strategy (STR).
55	Symbol	Y	Class root symbol for options or strategy symbol.
201	PutOrCall	N	Mandatory for Option.
202	StrikePrice	N	Mandatory for Option.
200	MaturityMonthYear	N	Mandatory for Option and Futures.

Tag	Field Name	Req	Comments
205	MaturityDay	N	Used in conjunction with MaturityMonthYear <200> to specify a particular maturity date.
206	OptAttribute	N	Indicates a corporate action
38	OrderQty	N	Number of shares ordered.
	Standard Trailer	Y	

4.8 b: Quote Acknowledgement

Tag	Field Name	Comments
	Standard Header	[MsgType 35=b]
131	QuoteReqID	Unique identifier for quote request.
297	QuoteAckStatus	Status of the quote acknowledgement. 0: Accepted 5: Rejected
300	QuoteRejectReason	Reason Quote was rejected.
	Standard Trailer	

4.9 c: Security Definition Request

This transaction can be used to:

- Request the creation of a New Instrument Series: either a Strategy or a Flexible Series (Option/Future).
- Query a list of all securities in the Exchange.

Technical Specifications

22 August 2014

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=c]
320	SecurityReqID	Y	Id for the security definition request.
321	SecurityRequestType	Y	1: Request New Instrument Series 2: Request List Security Types
167	SecurityType	Y	Only provided when [SecurityRequestType 321=1]. STR : Strategy OPT: Option FUT: Future
146	NoRelatedSym	Y	Only provided when [SecurityRequestType 321=1]. From 2 to 4 when SecurityType 167=STR: Strategy (Number of legs that make up the instrument) 1 when SecurityType 167=OPT: Option or SecurityType 167=FUT: Future
N Times (NoRelatedSym). The following fields are only provided when [SecurityRequestType 321=1].			
311	UnderlyingSymbol	Y	Underlying Symbol
310	UnderlyingSecurityType	Y	Security Type (required when SecurityType 167=STR: Strategy) OPT: Option FUT: Future
317	UnderlyingOptAttribute	N	Required if there is a corporate action on that leg
313	UnderlyingMaturityMonthYear	Y	Maturity Date



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Tag	Field Name	Req	Comments
314	UnderlyingMaturityDay	Y	Maturity Day
315	UnderlyingPutOrCall	N	Put or Call
316	Underlying Strike Price	N	Strike Price (mandatory for options)
319	RatioQty	Y	Ratio of the Strategy Leg (when SecurityType 167=STR only)
54	Side	Y	Defined as follows: 1: Buy 2: Sell
1194	OptionStyle	N	Default value is derived from Underlying 0: European 1: American.
	Standard Trailer	Y	

4.10d: Security Definition

The Security Definition message is the acknowledgement of a Security Definition Request message.

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=d]
320	SecurityReqID	Y	Unique ID of a Security Definition Request.
322	SecurityResponseID	Y	Unique ID of a Security Definition Message. Only present if [SecurityRequestType 321=2] in the Security Definition Request.

Tag	Field Name	Req	Comments
323	SecurityRequestType	Y	Only present if [SecurityRequestType 321=1] in the Security Definition Request. 1: Accept security proposal as is 2: Accept security proposal with revisions as indicated in the message 4: Already exists 5: Reject security proposal
55	Symbol	Y	Class root symbol for options or strategy symbol dependent on value of SecurityType. For Options/Futures: A Class Root Symbol (1-6 characters) For Strategies: External Symbol (30 character string)
167	SecurityType	Y	Indicates if Option (OPT), a Future (FUT), or Strategy (STR).
200	MaturityMonthYear		Provided for Options and Futures only, when SecurityType SecurityType 167 = OPT: Option or FUT: Future
205	MaturityDay		Provided for Options and Futures only, when SecurityType 167 = OPT: Option or FUT: Future
206	OptAttribute		Indicates a corporate action
201	PutOrCall		Provided for Options only, when SecurityType 167=OPT: Option

Tag	Field Name	Req	Comments
202	StrikePrice		Provided for Options only, when SecurityType 167=OPT: Option
1194	OptionStyle	N	For Options only. Default value is derived from Underlying 0: European 1: American.
146	NoRelatedSym	Y	From 2 to 40. Number of legs that make up the instrument. Provided only when SecurityType 167=STR: Strategy
N Times - The following fields are only provided only when [SecurityType 167=STR].			
311	UnderlyingSymbol	Y	Strategy Leg Symbol. See [Symbol 55] field for description.
310	UnderlyingSecurityType	Y	Strategy Leg SecurityType. See [SecurityType 167] field for description.
317	UnderlyingOptAttribute	Y	Strategy Leg corporate action indicator See [OptAttribute 206] field for description.
313	UnderlyingMaturityMonthYear	Y	Strategy Leg Maturity month and year. See [MaturityMonthYear 200] field for description.
314	UnderlyingMaturityDay	Y	Strategy Maturity day See [MaturityDay205] field for description.
315	UnderlyingPutOrCall	N	Strategy Leg PutOrCall. See [PutOrCall 201] field for description
316	UndelyingStrikePrice	N	Strategy Leg strike price. See [StrikePrice 202] field for description.
319	RatioQty	Y	Strategy Leg Ratio to determine valid quantity. Should not contain decimals
54	Side	Y	1: Buy 2: Sell
	Standard Trailer	Y	

Technical Specifications

22 August 2014

4.11j: Business Reject

Business reject [MsgType 35=J] is returned following a rejected Security Definition request [MsgType 35=c].

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=j]
45	RefSeqNum	Y	Message Seq Num for the 'j' message
372	RefMsgType	Y	The MsgType of the FIX message being referenced – This should be 'c' when strategy instrument cannot be created.
379	BusinessRejectRefID	Y	[SecurityReqID 320] from the submitted Security Definition request.
380	BusinessRejectReason	Y	Code to identify reason for reject: 0: Other 1: Unknown ID 2: Unknown Security (default value) 3: Unsupported Message Type 4: Application not available 5: Conditionally Required Field Missing
58	Text	Y	Free-form text to explain the reason for rejection.
	Standard Trailer	Y	

4.12s: New Order Cross

This message is used to enter a cross order.



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Technical Specifications

22 August 2014

Tag	Field Name	Req	Comments
	Standard Header	Y	[MsgType 35=s]
167	SecurityType	Y	STR: Strategy
55	Symbol	Y	Mandatory
201	PutOrCall	N	Mandatory for Option
202	StrikePrice	N	Mandatory for Option
200	MaturityMonthYear	N	Mandatory for Option and Future
205	MaturityDay	N	Mandatory for Equity Options
206	CorporateAction	N	Required if there is a corporate action
38	OrderQty	Y	Mandatory
44	Price	Y	Mandatory
552	NoSides	N	2: Two Sided
2 Times			
54	Side	Y	1: Buy 2: Sell
11	ClOrdID	Y	Unique identifier of the side (specified in Tag 54 above) as assigned by the institution
1	Account	Y	Account mnemonic as agreed between broker and institution.



London
Stock Exchange Group

Tag	Field Name	Req	Comments
8001	AccountProfile	N	H: Hedger S: Speculator
77	Open/Close	Y	O: Open C: Closed
58	Text	N	
47	Rule80A	Y	
	Standard Trailer	Y	

4.138: Execution Report

Execution Report can be generated for the following:

- An order gets traded
- A trade is cancelled by the Exchange on behalf of a participant
- An order gets eliminated
- An order gets expired
- A 'New Order - Single' confirmation
- An 'Order Cancel/Replace Request' confirmation
- An 'Order Cancel Request' confirmation
- In response to an 'Order Mass Status Request'
- An order gets updated by Self Execution Prevention (SEP)

Tag	Field Name	Comments
	Standard Header	[MsgType 35=8]
37	OrderID	Required to be unique for each chain of orders. Is set to "NONE" if in response to a New Order Single being rejected.

Tag	Field Name	Comments
11	ClOrdID	If provided in the original New Order, Cancel Request, or Order Cancel/Replace.
41	OrigClOrdID	Conditionally required for response to a Cancel or Cancel/Replace request [ExecType 150=Replaced or Cancelled]. [ClOrdID 11] of the previous order (NOT the initial order of the day) when cancelling or replacing an order. Not provided when [ExecTransType 20=1].
1	Account	Used for firm identification. If provided in the original order.
17	ExecID	Unique for each Execution Report message.
20	ExecTransType	See table below for possible values.
19	ExecRefID	Provided when [ExecTransType 20=1] or [ExecTransType 20=2]
150	ExecType	Describes the type of execution report. Possible values: 0: New 1: Partial fill 2: Fill 4: Cancelled 5: Replace 8: Rejected
103	OrdRejReason	May be provided only when [ExecType 150=8] (rejected).
39	OrdStatus	Describes the current state of a CHAIN of orders, same scope as OrderQty, [CumQty 14], [LeavesQty 151], and [AvgPx 6]. See table below for possible values.
38	OrderQty	Quantity of the order (as opposed to [LastShares 32] which refer to the fill).
44	Price	Price of the order (as opposed to [LastPx 31] which refer to the fill). If provided in the original order.
167	SecurityType	Indicates if Option (OPT), a Future (FUT), or Strategy (STR).
55	Symbol	Class root symbol for options or strategy symbol.
201	PutOrCall	For Options only.
202	StrikePrice	For Options only.

Tag	Field Name	Comments
200	MaturityMonthYear	For Options and futures only.
205	MaturityDay	Used in conjunction with MaturityMonthYear <200> to specify a particular maturity date.
206	OptAttribute	Indicates a corporate action
210	MaxShow	Maximum number of shares within an order to be shown to other customers
54	Side	Buy or Sell
40	OrdType	Please refer to Field Definitions.
59	TimeInForce	Absence of this field indicates Day order
432	ExpireDate	Provided when [TimeInForce 59=6] (GTD)
32	LastShares	Quantity of shares bought/sold on this (last) fill.
31	LastPx	Price of this (last) fill.
151	LeavesQty	Amount of shares open for further execution. If the [OrdStatus 39=4] (Cancelled), then [LeavesQty 151] could be 0, otherwise [LeavesQty 151] = [OrderQty 38] - [CumQty 14].
14	CumQty	Currently executed shares for chain of orders.
6	AvgPx	Calculated average price of all fills on this order.
60	TransactTime	Time of execution/order creation, expressed in UTC.
77	Open/Close	Please refer to Field Definitions.
442	MultipleReportingType	Indicates the type of instrument the Execution Report refers to. Used with multi-leg securities such as, Options Strategies, Spreads, etc.
10455	SecurityAltID	Indicates the Strategy Instrument in Execution Report Message sent for the Strategy. Send only when MultipleReportingType 442 = 2 Example: If 35 = 8 and 442 = 2, then 10455 = Symbol of the Strategy.
527	SecondaryExecID	Use to refer to the Strategy ExecID. Send only when MultipleReportingType 442 = 2 Example: If 35 = 8 and 442 = 2, then 527 = ExecID of the Strategy.

Tag	Field Name	Comments
58	Text	If provided in the original order. When [ExecType 150=8] (rejected), contains the text associated with the reason of the rejection.
47	Rule80A	Identifies the type of account.
382	NoContraBrokers	Value will always be 1 when OrdStatus (Tag 39) equals 1 or 2.
375	ContraBroker	Value will be equal to the opposite Broker ID of the trade when OrdStatus (Tag 39) is equal to 1 or 2.
584	MassStatusReqID (taken from FIX 4.4)	Required if responding to a Order Mass Status Request. Echo back the value provided by the requester.
337	Contra Trader	Returned when tag is sent for a Committed Order
5255	StopPxCondition	Returned when tag is sent for a Stop Order
9459	SpecialTradeInd	Returned for a trade message
84	CxlQty	Quantity removed by Self Execution Prevention (SEP)
	Standard Trailer	

The following table defines the use of fields, [ExecType 150], [OrdStatus 39], [ExecTransType 20] according to the action performed on an Order or a Trade (Fill).

Action	[ExecType 150]	[OrdStatus 39]	[ExecTransType20]
New Order accepted and put on book	0: New	0: New	0: New
New Order already received with [PossResend 97=Y]	According to the status of the order at that time	According to the status of the order at that time	3: Status

Action	[ExecType 150]	[OrdStatus 39]	[ExecTransType20]
A Stop Order is triggered and transformed into a regular order. In this case, the [OrdType 40] is changed to 2 (limit).	0: New	0: New	0: New
Response to an 'Order Mass Status Request' message [MsgType 35=AF]	The current status of the order. 0: New 1: Partial fill 2: Fill	The current status of the order. 0: New 1: Partial fill 2: Fill	3: Status
Order Traded	1: Partial fill 2: Fill	1: Partial filled 2: Filled	0: New
Order Eliminated for one of the following reasons: • by the Exchange • by the General Clearing Member (GCM) • because of an update on the instrument • an 'Immediate or Cancel' order is not filled or partially filled • a MinQty order that cannot be filled immediately for the minimum specified quantity when it is received by the Exchange	4: Cancelled	4: Cancelled	0: New
Order Eliminated by Self Execution Prevention	4: Cancelled	5: Eliminated by SEP	0: New
Order Cancel/Replace accepted and put in the book	5: Replace	5: Replaced	0: New
Order Cancel/Replace accepted and put in the book on a partially filled order	5: Replace	1: Partial filled	0: New
Order Expired	C: Expired	C: Expired	0: New
Trade Killed (Busted)	1: Partial fill 2: Fill	The current status of the order. 0: New 1: Partial fill 4: Cancelled	1: Cancel

4.149: Order Cancel Reject

Tag	Field Name	Comments
	Standard Header	[MsgType 35=9]
37	OrderID	If the order is unknown the OrderID (37) will equal 'NONE', otherwise the OrderID (37) will be populated with the OrderID (37) of the order referenced in the Cancel or Cancel Replace message.
11	ClOrdID	Unique order ID assigned by institution to the cancel request or to the replacement order.
41	OrigClOrdID	[ClOrdID 11] which could not be cancelled/replaced. [ClOrdID 11] of the previous order (NOT the initial order of the day) when cancelling or replacing an order.
39	OrdStatus	Value after this cancel reject is applied.
434	CxlRejResponseTo	Identifies the type of request that a Cancel Reject is in response to.
58	Text	Contains the reason why the Cancel/Replace message has been rejected.
	Standard Trailer	

5.0 Field Definitions

Tag	Field Name	Description	Field Length
1	Account	Account mnemonic as agreed between broker and institution.	1 to 12
6	AvgPx	Calculated average price of all fills on this order.	1 to 9
7	BeginSeqNo	Message sequence number of first message in range to be resent	1 to 9
8	BeginString	Identifies beginning of new message and protocol version. ALWAYS FIRST FIELD IN MESSAGE. Valid values: FIX.4.2	7
9	BodyLength	Message length, in bytes, forward to the [Checksum 10] field. ALWAYS SECOND FIELD IN MESSAGE.	2 to 4
10	Checksum	Three bytes, simple checksum (see FIX 4.2 document for description). ALWAYS LAST FIELD IN MESSAGE; i.e. serves, with the trailing <SOH>, as the end-of-message delimiter. Always defined as three characters.	3
11	ClOrdID	Unique identifier for Order as assigned by institution. Uniqueness must be guaranteed within a single trading day. Firms which submit multi-day orders should consider embedding a date within this field to assure uniqueness across days.	1 to 50
14	CumQty	Total number of shares filled.	1 to 9
16	EndSeqNo	Message sequence number of last message in range to be resent. If request is for a single message [BeginSeqNo 7 = EndSeqNo 16]. If request is for all messages subsequent to a particular message, [EndSeqNo 16=0].	1 to 9
17	ExecID	Unique identifier of execution message as assigned by the Exchange. It will be 0 (zero) for [ExecTransType 20=3] (Status).	1 to 15
19	ExecRefID	Reference identifier used with Cancel transaction types.	1 to 15

Tag	Field Name	Description	Field Length
20	ExecTransType	Identifies transaction type. Valid values: 0: New 1: Cancel 3: Status	1
31	LastPx	Price of this (last) fill.	1 to 9
32	LastShares	Quantity of shares bought/sold on this (last) fill	1 to 9
34	MsgSeqNum	Message sequence number.	1 to 9
35	MsgType	Defines message type. ALWAYS THIRD FIELD IN MESSAGE Valid values: 0: Heartbeat 1: Test Request 2: Resend Request 3: Reject 4: Sequence Reset 5: Logout 8: Execution Report 9: Order Cancel Reject A: Logon D: New Order - Single F: Order Cancel Request G: Order Cancel/Replace Request R: Quote Request b: Quote Acknowledgement c: Security Definition Request d: Security Definition AF: Order Mass Status Request s: New Order Cross	2
36	NewSeqNo	New sequence number	1 to 9
37	OrderID	Unique identifier for Order as assigned by the Exchange	10 to 20
38	OrderQty	Number of shares ordered.	1 to 9

Tag	Field Name	Description	Field Length
39	OrdStatus	Identifies current Status of order. Valid values: 0: New 1: Partially filled 2: Filled 4: Cancelled 5: Replaced 8: Rejected F: Eliminated out of Limits G: Eliminated by Circuit Breaker M: Eliminated by Market Control I: Eliminated on Disconnect U: Eliminated due to Unpriced Leg Z: Eliminated by Self Execution Prevention (SEP)	1
40	OrdType	Indicates the Price Type of the order. Must contain one of the following values for regular orders: 1: Market Order: Top of book order (only executes against best opposite side offer) 2: Limit 3: Stop Market (top of book) 4: Stop C: Committed V: U.S. Market Order (executes against different levels of the book) W = Stop U.S. Market	1
41	OrigClOrdID	[ClOrdID 11] of the previous order (NOT the initial order of the day) as assigned by the institution, used to identify the previous order in cancel and cancel/replace requests	1 to 50
43	PossDupFlag	Indicates possible retransmission of message with this sequence number Valid values: Y: Possible duplicate N: Original transmission	1
44	Price	Price per contract.	1 to 9
45	RefSeqNum	Reference message sequence number	1 to 9

Tag	Field Name	Description	Field Length
47	Rule80A	Identify the Account Type. Valid values: N: Non-Segregated Client P: Market Maker F: House C: Client	1
49	SenderCompID	Assigned value used to identify the sender in a FIX session	4 to 8
50	SenderSubID	Assigned value used to identify specific message originator (desk, trader, etc.). Participant can have this tag configured on the Exchange Participant database to enable them to use a single FIX session to send orders for different Traders/Users.	Up to 11 characters
52	SendingTime	Time of message transmission (always expressed in UTC)	17 to 21
54	Side	Verb of order. Valid values: 1: Buy 2: Sell	1
55	Symbol	Class root symbol for options or strategy symbol	1 to 30
56	TargetCompID	Assigned value used to identify the receiver in a FIX session.	4 to 8
57	TargetSubID	Assigned value used to identify specific individual or unit intended to receive message. "ADMIN" reserved for administrative messages not intended for a specific user. If tag 50 is present in an incoming message, then the response will contain tag 57 which will return the same value as tag 50.	Up to 11 characters

Tag	Field Name	Description	Field Length
58	Text	Free format text string. If provided in the original order. When [ExecType 150=8] (rejected), contains the text associated with the reason of the rejection.	1 to 18 in New Order message 1 to 80 otherwise
59	TimeInForce	Specifies how long the order remains in effect. Absence of this field is interpreted a DAY. Valid values: 0: Day 1: Good Till Cancel (GTC) 3: Immediate or Cancel (IOC) 6: Good Till Date W: WhileConnected	1
60	TransactTime	Time of execution/order creation, expressed in UTC	17 to 21

Tag	Field Name	Description	Field Length
77	Open/Close	This data field indicates how the participant's position will be handled by the clearing system. Must contain one of the following values: Any number of Legs or Single Security: O: All Legs are open or single security C: All Legs are closed or single security 2 Legged Strategy: 1: 1st Leg Open, 2nd Leg Close 2: 1st Leg Close, 2nd Leg Open 3 Legged Strategy: 3: 1st Leg Open, 2nd Leg Open, 3rd Leg Close 4: 1st Leg Open, 2nd Leg Close, 3rd Leg Open 5: 1st Leg Open, 2nd Leg Close, 3rd Leg Close 6: 1st Leg Close, 2nd Leg Open, 3rd Leg Open 7: 1st Leg Close, 2nd Leg Open, 3rd Leg Close 8: 1st Leg Close, 2nd Leg Close, 3rd Leg Open 4 Legged Strategy: A: 1st Leg Open, 2nd Leg Open, 3rd Leg Open, 4th Leg Close B: 1st Leg Open, 2nd Leg Open, 3rd Leg Close, 4th Leg Open D: 1st Leg Open, 2nd Leg Open, 3rd Leg Close, 4th Leg Close E: 1st Leg Open, 2nd Leg Close, 3rd Leg Open, 4th Leg Open F: 1st Leg Open, 2nd Leg Close, 3rd Leg Open, 4th Leg Close G: 1st Leg Open, 2nd Leg Close, 3rd Leg Close, 4th Leg Open H: 1st Leg Open, 2nd Leg Close, 3rd Leg Close, 4th Leg Close I: 1st Leg Close, 2nd Leg Open, 3rd Leg Open, 4th Leg Open J: 1st Leg Close, 2nd Leg Open, 3rd Leg Open, 4th Leg Close K: 1st Leg Close, 2nd Leg Open, 3rd Leg Close, 4th Leg Open L: 1st Leg Close, 2nd Leg Open, 3rd Leg Close, 4th Leg Close M: 1st Leg Close, 2nd Leg Close, 3rd Leg Open, 4th Leg Open N = 1st Leg Close, 2nd Leg Close, 3rd Leg Open, 4th Leg Close P = 1st Leg Close, 2nd Leg Close, 3rd Leg Close, 4th Leg Open	1

Tag	Field Name	Description	Field Length
84	CxlQty	Indicates the quantity removed by Self Execution Prevention (SEP)	1 to 9
97	PossResend	Indicates that message may contain information that has been sent under another sequence number. Valid Values: Y: Possible resend N: Original transmission	1
98	EncryptMethod	Method of encryption. Valid values: 0: None	1
99	StopPx	For on Stop order (Stop limit only). Required when [OrdType 40 = 4]	1 to 9
103	OrdRejReason	Code to identify reason for order rejection. 0: Broker option 1: Unknown symbol 2: Exchange closed 3: Order exceeds limit 4: Too late to enter 5: Unknown Order 6: Duplicate Order (e.g. dupe ClOrdID) 7: Duplicate of a verbally communicated order 8: Stale Order 9: Group state doesn't allow this function	1 to 6
108	HeartBtInt	Heartbeat interval (seconds). Must be equal or greater than 30 or equal to 0 (no heartbeat)	1 to 4
110	MinQty	Minimum quantity of an order to be executed.	1 to 9
112	TestReqID	Identifier included in Test Request message to be returned in resulting Heartbeat	1 to 20
122	OrigSendingTime	Original time of message transmission (always expressed in UTC) when transmitting orders as the result of a resend request.	17 to 21

Tag	Field Name	Description	Field Length
123	GapFillFlag	Indicates that the Sequence Reset message is replacing administrative or application messages which will not be resent. Valid values: Y: Gap Fill message, MsgSeqNum field valid N: Sequence Reset, ignore MsgSeqNum	1
131	QuoteReqID	Unique identifier for quote request	1 to 50
141	ResetSeqNumFlag	Indicates that the both sides of the FIX session should reset sequence numbers Valid values: Y: Yes, reset sequence numbers N: No	1
146	NoRelatedSym	Specifies the number of repeating symbols specified.	1
150	ExecType	Describes the specific Execution Report. Valid values: 0: New 1: Partial fill 2: Fill 4: Cancelled 5: Replace 8: Rejected F: Trade (partial fill or fill)	1
151	LeavesQty	Amount of shares open for further execution.	1 to 9

Tag	Field Name	Description	Field Length
167	SecurityType	Indicates the type of security. Valid values: OPT: Option FUT: Future STR: Strategy For Options: [Symbol 55], [PutOrCall 201], [MaturityMonthYear200] and [StrikePrice 202] are required. For Futures: [Symbol 55], and [MaturityMonthYear200] are required. For Strategy: only [Symbol 55] is required.	3
200	MaturityMonthYear	Month and Year of the maturity for [SecurityType 167 = OPT].	6
201	PutOrCall	Indicates whether an Option is for a put or call. Valid values: 0: Put 1: Call 2: Over 3: Under	1
202	StrikePrice	Strike Price for an Option. A maximum of 4 decimals are accepted.	1 to 9
205	MaturityDay	Used in conjunction with MaturityMonthYear <200> to specify a particular maturity date.	1 to 2
206	OptAttribute	Indicates a corporate action (Supported Values: X, Y and Z)	1
210	MaxShow	Maximum number of shares within an order to be shown to other customers	1 to 9
297	QuoteAckStatus	Identifies the status of the quote acknowledgement. Valid values: 0: Accepted 5: Rejected	1

Tag	Field Name	Description	Field Length
300	QuoteRejectReason	Reason Quote was rejected. See Error Code Section for the list of error codes	4 to 6
310	UnderlyingSecurityType	Strategy leg SecurityType. See [SecurityType 167] field for description	3
311	UnderlyingSymbol	Strategy leg Symbol. See [Symbol 55] field for description	1 to 30
313	UnderlyingMaturity MonthYear	Strategy leg maturity date. See [MaturityMonthYear 200] field for description.	6
314	UnderlyingMaturityDay	Strategy Leg maturity day. See [MaturityDay 205] field for description.	1 or 2
315	UnderlyingPutOrCall	Strategy leg PutOrCall. See [PutOrCall 201] field for description	1
316	UndelyingStrikePrice	Strategy leg strike price. See [StrikePrice 202] field for description.	1 to 9
317	UnderlyingOptAttribute	Strategy Leg corporate action indicator. See [OptAttribute 206] field for description.	1
319	RatioQty	Strategy leg ratio to determine a valid quantity. Should not contain decimals	1 to 9
320	SecurityReqID	Unique ID of a Security Definition Request.	1 to 50
321	SecurityRequestType	Type of Security Definition Request.	1
322	SecurityResponseID	Unique ID of a Security Definition message.	1 to 15
371	RefTagID	The tag number of the FIX field being referenced.	1 to 5
372	RefMsgType	The message type of the FIX message being referenced.	1 to 2
373	SessionRejectReason	Code to identify reason for a session-level Reject message.	1 to 2

Tag	Field Name	Description	Field Length
375	ContraBroker	Exchange number of the firm on the opposite side of the trade. Value will be equal to the receiver's Exchange number when reporting cross trades. For all other messages, tag [ContraBroker 375] will not be supplied.	3
382	NoContraBrokers	Number of ContraBrokers repeating group instances. Always appear before tag [ContraBroker 375]. Value will be 1 when reporting cross trades. For all other messages, tag [NoContraBroker 382] will not be supplied.	1
383	MaxMessageSize	Maximum number of bytes supported for a single message.	2 to 4
432	ExpireDate	Date of order expiration (last day the order can trade), always expressed in terms of the local market date	8
434	CxlRejResponseTo	Identifies the type of request that a Cancel Reject is in response to. Valid values: 1: Order Cancel Request 2: Order Cancel/Replace Request	1
442	MultipleReportingType	Indicates the type of instrument the Execution Report refers to. Used with multi-leg securities such as, Options Strategies, Spreads, etc. Valid Values are: 1: Single security or outright contract(default) 2: Individual leg of a multi-leg or Strategy Instrument 3: Multi-leg or Strategy Instrument	1
448	PartyID	Executing Firm ID	4

Tag	Field Name	Description	Field Length
527	SecondaryExecID	Used to refer to the Strategy ExecID. Send only when MultipleReportingType 442 = 2 Example: If 35 = 8 and 442 = 2, then 527 = ExecID of the Strategy	1 to 20
555	NoLegs	Number of legs	1
566	LegPrice	Order price	1 to 9
571	TradeReportID	Proposal ID Only for messages coming from Exchange	8
577	ClearingInstruction	Clearing instruction	12
581	CaptRptAccountType	Identify the Account Type. Valid values: N: Non-Segregated Client P: Market Maker F: House C: Client	1
584	MassStatusReqID	This field is taken from FIX 4.3 Value assigned by issuer of Mass Status Request to uniquely identify the request.	1 to 50
585	MassStatusReqType	This field is taken from FIX 4.3. Mass Status Request Type Valid values: 7 = Status for all orders	1

Tag	Field Name	Description	Field Length
600	LegSymbol	See [Symbol 55] field for description.	1 to 30
609	LegSecurityType	See [SecurityType 167] field for description.	3
610	LegMaturityMonthYear	See [MaturityMonthYear 200] field for description.	6
612	LegStrikePrice	. See [StrikePrice 202] field for description.	1 to 9
613	LegOptAttribute	See [OptAttribute 206] field for description.	1
624	LegSide	Side of the order. 1: Buy 2: Sell	1
687	LegQty	Order quantity	1 to 9
852	PublishTrdIndicator	Transparency for the Flexible trade. Y: Published N: Unpublished	1
856	TradeReportType	2: Accept 3: Decline	1
1041	FirmTradeID	The ID assigned to a trade by the Firm to track a trade within the Firm system.	8
1117	RootPartyID	Root Party ID	4

Tag	Field Name	Description	Field Length
1194	OptionStyle	Valid for options only. 0: European 1: American.	1
1358	LegPutOrCall	See [PutOrCall 201] field for description.	1
5255	StopPxCondition	Required if Tag 99 (StopPx) is filled. Indicates the type of stop orders. S: Regular Stop (optional: default value used if none provided) T: If Touched E: Stop On Bid F: If Bid Touched I: Stop On Ask H: If Ask Touched	1
8001	AccountProfile	Indicate the type of trader. Valid Values: H: Hedger S: Speculator (default value if not provided in the order)	1
9433	TextExecutionReport	Free text	50
9459	SpecialTradeInd	Indicates the type of trade. ' ': Regular Trade A: AsOfTrade L: LateTrade 1: ExchangeGranted1 B: BlockTrade K: CommittedBlock T: Committed D: Cross For clients using an older FIX version, this tag will not be sent.	

Tag	Field Name	Description	Field Length
10455	SecurityAltID	Indicates the Strategy Instrument in Execution Report Message sent for the Strategy. Send only when MultipleReportingType 442 = 2 Example: If 35 = 8 and 442 = 2, then 10455 = Symbol of the Strategy.	1 to 15
16453	ProposalType	B: Bundle Order T: Inter Dealer Broker Order	1
16454	LegMaturityDay	Leg Maturity Day. See [MaturityDay 205] field for description.	1 or 2

6.0 Error Codes

The following table displays the error codes and text that will appear in error responses.

Error Code	Error Description
1	User Identification is not correct
2	Protocol Version is not supported
3	Message Type is not supported
4	Session ID is not active
6	Message Type requested is not supported
8	Message is too short
9	Message is too long
10	Message contains Binary Data
11	No Heartbeat Activity: Disconnection
12	Message Type is Out Of Context
13	User ID has been deactivated: Disconnection
14	Syntax Error + <detailed text>
15	Field value is too small
16	Field value is too big
100	Firm is Forbidden to trade on this Group
101	Duration Type is Forbidden for current Group state
102	Verb field (Side) cannot be modified
103	Order is not active
104	Price Type is forbidden for this instrument
105	Price Term is Forbidden for current Instrument state
108	Duration Type is Forbidden for current Instrument state
109	Order cannot be processed: No opposite limit
110	Price does not represent a valid tick increment for this Instrument
111	Duration Type is invalid for this Price Type
112	Cross Order price must be within the Instrument trading limits
113	Cross Order price is outside bid/ask price spread
114	Opposite firm must be filled for committed order
116	Cross order is not allowed
117	Cross order quantity is outside limits
118	Duration Type Is Invalid For This Price Term

Error Code	Error Description
201	GTD date must be equal to or greater than current day
202	GTD date must be equal to or less than Instrument expiration date
203	GTD date must be filled only if Duration Type is equal to GTD
300	Quantity Term is Forbidden for current Instrument state
302	Quantity must be less than or equal to Maximum Improvement Quantity
303	Quantity Term is not authorized for this Order Type
304	Additional Quantity must be less than Order Quantity
305	Additional Quantity is too small
306	Minimum quantity cannot be modified
307	Quantity Term is forbidden for Duration Type
308	Order quantity is outside the instrument quantity threshold
309	Quantities must be multiples of lot size
402	Trader ID field cannot be modified
403	Market Maker not authorized for Group
500	Order price is outside the instrument price threshold
501	Price field is mandatory for Limit Orders
502	Price field must not be filled for this Price Type
503	Order cannot be modified or cancelled
504	Additional Price is forbidden for Price Term
505	Order price must be greater than additional price
506	Order price must be lower than additional price
507	Additional price must be lower than last price or last day price
508	Additional price must be greater than last price or last day price
509	Order rejected. Cannot trade outside instrument price thresholds.
510	Order cannot be modified
511	Order price is outside circuit breaker limits
512	Price Term Invalid For Price Type
700	Only one quote per Instrument and per Side is accepted
701	Quote is not present in the Instrument Book
702	Market Maker Protection in progress; Quote not processed.
703	Advanced Market Maker Protection not enabled for this Group
704	Buy and Sell must not cross for the same instrument
705	Number of quotes is not in sync with the message length
707	Market Maker Protection state must be re-activated

Error Code	Error Description
708	Trader ID has no quote for this Group
709	All the Instruments must belong to the same Group
710	Clearing Data has not been initialized
1000	Cross orders forbidden in Pre-opening phase.
1001	Instrument does not exist
1002	Group ID does not exist
1003	Trader ID is invalid
1004	Message Type is forbidden for current Instrument state
1007	Participant must use A4 protocol version
1008	RFQ currently underway for this instrument
1009	Action not allowed under current configuration
1010	Number of entries is invalid
1107	Firm or trader had been disabled
1108	Instrument mandatory when using MM Monitoring mode forced
1109	Market maker has no obligation for this group
1110	Participant not authorized for this Group (for message 35 = 8)
1111	Participant not authorized for this Account Type (for message 35 = 8)
2000	Technical error; function not performed. Contact Technical Help Desk.
2001	Gateway State forbids this command. Contact Technical Help Desk.
2002	Function only partially performed. Contact Technical Help Desk.
3017	Open Close Missing Invalid
3041	Unknown Clearing Operation Mode
3042	Invalid Price Type
9017	Invalid number of legs
9018	Invalid leg information
9019	Unknown strategy type
9020	Firm creation quota has been reached
9021	Leg instrument is not active
9022	Strategy has unpriced legs
9023	Group state does not allow this function
9024	Legs have different Multi-group Strategy Key
9025	Legs have different Multi-group Strategy Group
9026	Order rejected. Cannot assign a valid price to all legs
9027	Maximum pending flexible creation reached

Error Code	Error Description
9028	Duration type is invalid for this Message Type
9029	Legs must be on the same CPU
9030	Strike price is not multiple of tick size
9031	Symbol Date is not a valid trading day
9032	Symbol Date is out of limits
9033	Invalid Proposal ID or Confirmation Order ID
9034	Invalid order type on Flexible Instrument
9035	Trading not allowed on Flexible Instrument
9036	Bundle creation quotas has been reached for the Firm
9037	Proposal creation quotas has been reached for the Firm

7.0 Message Flow

7.1 Connection Management

Establishing a FIX Session

In order to establish a FIX session, clients need to pay special attention to the following items:

- Clients are allowed to establish FIX sessions on each trading day
- The Exchange CompID is 'LSE1'
- No encryption technique is supported
- For a Resend Request, the Exchange will always request all messages subsequent to the last received ([BeginSeqNo 7=first message of range], [EndSeqNo 16=0]). Any new messages received before the reception of the last requested message will be discarded
- The Exchange does not support 24-hour connectivity and a new session must be re-established each morning
- The Exchange allows only one participant per FIX session (or per CompID). ISVs offering a service bureau are required to have a separate CompID, and therefore a separate FIX session, for each of their clients
- For message traffic efficiency, the Exchange does not allow clients to send a Heartbeat message at intervals of less than 30 seconds. Consequently, Logon messages with a value of less than 30 seconds, and not equal to 0 (meaning no Heartbeat will be sent) in the [HeartBtInt 108] field are rejected. If a participant does not send a message within 2 Heartbeat periods, the Exchange will send a Test Request message and wait for a response from the participant. If the participant does not respond to the Test Request within the Heartbeat

period, the session will be disconnected. To summarize, if a participant does not send any messages within 3 Heartbeat periods, the connection is dropped and the participant will need to reconnect.

Initializing a FIX Connection

PARTICIPANT		EXCHANGE
Logon [MsgType 35 = A]	→	
	←	Logon [MsgType 35 = A]

Terminating a FIX Connection

PARTICIPANT		EXCHANGE
Logout [MsgType 35 = 5]	→	
	←	Logout [MsgType 35 = 5]

Sending a Heartbeat

PARTICIPANT		EXCHANGE
Heartbeat [MsgType 35 = 0]	→	

Heartbeat Management

PARTICIPANT		EXCHANGE
Test Request [MsgType 35 = 1] [TestReqID 112 = value]	→	
	←	Heartbeat [MsgType 35=0] [TestReqID 112 = value]

Sending a Resend Request

PARTICIPANT	EXCHANGE
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PARTICIPANT		EXCHANGE	
Resend Request [MsgType 35 = 2] [BeginSeqNo 7 = seqno] [EndSeqNo 16 = seqno or 0]	→		
	←	Resend Original Messages [PossDupFlag 43 = Y] and/or [MsgType 35 = 4] [ResetSeqNumFlag 141 = Y]	

Sending a Sequence Reset

PARTICIPANT		EXCHANGE	
Sequence Reset [MsgType 35 = 4] [ResetSeqNumFlag 141 = Y]	→		
	←	Resend Original Messages [PossDupFlag 43 = Y] or Sequence Reset [MsgType 35 = 4] [ResetSeqNumFlag 141 = Y]	

Receiving a Session Level Reject

PARTICIPANT		EXCHANGE	
New Order [MsgType 35 = D] required field missing	→		
	←	Reject [MsgType 35 = 3] [RefSeqNum 45 = offending msg#] [Text 58 = description]	

Using a Single FIX Session to Send Orders for Multiple Traders/Users

Participants may use a single FIX session to send orders for different Traders/Users. In order to utilise this functionality the following optional tags are available on the FIX interface:

- Tag 50: SenderSubID, alphanumeric, maximum 11 characters
- Tag 57: TargetSubID, alphanumeric, maximum 11 characters

Tags 50 and 57 are commonly used as a unique value to identify a specific message originator, for example a specific trading desk or trader. For all FIX message types, the Exchange will support tag 50 SenderSubID and return back the same value in tag 57 TargetSubID. These tags will be part of the Standard Message Header.

As with the CompIDs, the Exchange will map SubIDs to TraderIDs. Values used by participants in tag 50 will need to be configured in the Exchange participant database. All messages received with a value not configured will still be accepted but will be associated to the default TraderID configured for the CompID. Messages received without tag 50 will use the default TraderID as well (see following table). Consequently, this change is transparent for Participants who do not wish to implement and support this change.

Note: Participants who choose to support these new tags will need to contact the Exchange to have their new SubIDs configured.

Scenarios	Client CompID	Default TraderID linked with CompID	Client SubID	TraderID linked with SubID (new configuration)	TraderID used on the Exchange
Client SubID not provided	ABCDE	999S0F1	None	None	999S0F1
Client SubID provided but not configured on the Exchange	ABCDE	999S0F1	2-ABC	999S0Q1	999S0F1
Client SubID provided and configured on the Exchange	ABCDE	999S0F1	3-ABC	999S0Q1	999S0Q1

7.2 Order Processing

To enter an order, a participant sends a NEW ORDER SINGLE message.

The Exchange carries out validation on the parameters of the message received. If one of these parameters is invalid, the Exchange sends back an EXECUTION REPORT - REJECT rejecting the message received and specifying the first error detected.

If validation is successful, the Exchange accepts the message received and attributes an [OrderId 37] to the order entered.

An EXECUTION REPORT message received immediately after an order entry can indicate that it has been:

- Entered on the order book (a part of the order having possibly been executed)
- Eliminated
- Executed partially or in full
- Rejected

If an order is either partially or fully executed, the client receives, immediately after the EXECUTION REPORT ([OrdStatus 39=0]) message, one or more EXECUTION REPORT ([OrdStatus 39=1 or 2]) messages providing more information about the trade that took place.

If the order was on a strategy instrument, the client receives an EXECUTION REPORT for the strategy instrument ([OrderStatus 39=1 or 2, MultiLegReportingType = 3]) and additional EXECUTION REPORT messages for each leg ([OrderStatus 39=1 or 2, MultiLegReportingType = 2]) providing additional information related to the price and quantity at which each of the individual legs of the strategy instrument traded.

If the order has been booked (an EXECUTION REPORT message is sent with [OrdStatus 39=0]) the participant will automatically receive at a later time one of the following messages:

- One or more EXECUTION REPORT ([OrdStatus 39=1 or 2]) messages
- In the case of a Strategy Order, several EXECUTION REPORT messages ([OrderStatus 39=1 or 2, MultiLegReportingType 442=2]) in addition to the EXECUTION REPORT on the strategy instrument ([OrderStatus 39=1 or 2, MultiLegReportingType 442=3]) are sent. Each of the leg EXECUTION REPORT can be linked to its parent strategy trade (EXECUTION REPORT on the strategy instrument) message by the Strategy Instrument Id ([SecurityAltId 10455]) and Strategy Execution Id ([SecondaryExecId 527]).
- An EXECUTION REPORT ([OrdStatus 39=4]) message if the order is eliminated

Order is rejected

PARTICIPANT		EXCHANGE
New Order [MsgType 35 = D]	→	
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 8] [OrdStatus 39 = 8] [ExecTransType 20 = 0]

Order is accepted and fully executed

PARTICIPANT		EXCHANGE
New Order [MsgType 35 = D]	→	
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 0] [OrdStatus 39 = 0] [ExecTransType 20 = 0]
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 1] [OrdStatus 39 = 1] [ExecTransType 20 = 0]
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 2] [OrdStatus 39 = 2] [ExecTransType 20 = 0]

Minimum Quantity Order is not executed when entered

PARTICIPANT		EXCHANGE
New Order [MsgType 35 = D] [MinQty 110 = 20]	→	
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 4] [OrdStatus 39 = 4] [ExecTransType 20 = 0]

New Order is sent with Possible Resend

PARTICIPANT	EXCHANGE
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PARTICIPANT		EXCHANGE
New Order [MsgType 35 = D] [PossResend 97 = Y]	→	
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 4] [OrdStatus 39 = 4] [ExecTransType 20 = 0]

Order is resent with Possible Resend

PARTICIPANT		EXCHANGE
New Order [MsgType 35 = D] [PossResend 97 = Y]	→	
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 0] [ExecTransType 20 = 0]

Stop and If Touched Order

If the order entered is a STOP or IF TOUCHED order, the participant will later receive another EXECUTION REPORT message with [OrdStatus 39 = 0] and [OrdType 40 = 2] when the order is triggered and becomes a regular limit order (trigger price reached).

By design, it's not possible to enter Stop orders with Duration Type FAK but just Day or Market.

The following table illustrates all possible triggering surfaces:

Triggering Surface	Verb	StopPxCondition
Trigger Price $\geq$ LAST	Buy	5255 = T: If Touched
	Sell	5255 = S: Stop
Trigger Price $\leq$ LAST	Buy	5255 = S: Stop
	Sell	5255 = T: If Touched
Trigger Price $\geq$ BID	Buy	5255 = F: If BID Touched
	Sell	5255 = E: Stop on BID

Triggering Surface	Verb	StopPxCondition
Trigger Price ≤ BID	Buy	5255 = E: Stop on BID
	Sell	5255 = F: If BID Touched
Trigger Price ≥ ASK	Buy	5255 = H: If ASK Touched
	Sell	5255 = I: Stop on ASK
Trigger Price ≤ ASK	Buy	5255 = I: Stop on ASK
	Sell	5255 = H: If ASK Touched

Stop Order is triggered

PARTICIPANT	EXCHANGE		
New Order [MsgType 35 = D] [StopPx 99 = 1.00]	→		
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 0] [OrdStatus 39 = 0] [ExecTransType 20 = 0] [OrdType 40 = 4]	
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 0] [OrdStatus 39 = 0] [ExecTransType 20 = 0] [OrdType 40 = 2]	<i>When the Stop order is triggered and becomes a regular limit order.</i>

Immediate Order Completely Filled upon entry

Immediate or Cancel order (IOC) is also referred to as a Fill and Kill order. The participant will receive an EXECUTION REPORT ([OrdStatus 39=0]) message indicating that the order has been accepted. Then one or more EXECUTION REPORT ([OrdStatus 39=1 or 2]) messages will be sent. In each of the reports, the [LeavesQty 151] field specifies whether a part of the order remains to be traded. In the last EXECUTION REPORT, the [LeavesQty 151] is set to 0.

In the case of a strategy order fill, several EXECUTION REPORT messages ([OrdStatus 39=1 or 2, MultiLegReportingType 442=2]) in addition to the EXECUTION REPORT on the strategy instrument are sent.

Immediate Order partially Executed in Continuous Trading

PARTICIPANT	EXCHANGE
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PARTICIPANT		EXCHANGE
New Order [MsgType 35 = D] [TimeInForce 59 = 3]	→	
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 0] [OrdStatus 39 = 0] [ExecTransType 20 = 0]
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 1] [OrdStatus 39 = 1] [ExecTransType 20 = 0]
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 4] [OrdStatus 39 = 4] [ExecTransType 20 = 0]

Immediate is not Executed in Continuous Trading

PARTICIPANT		EXCHANGE
New Order [MsgType 35 = D] [TimeInForce 59 = 3]	→	
	←	Execution Report [MsgType 35 = 8] [ExecType 150 = 4] [OrdStatus 39 = 4] [ExecTransType 20 = 0]

Fill or Kill functionality can be simulated using a combination of TimeInForce (59 = 3) and MinQty (110 = <total quantity>).

7.3 Time Validity

Order designated as "While Connected" [TimelnForce 59 = W] is automatically cancelled following a participant disconnection.

While Connected orders cancelled on disconnection of a participant

PARTICIPANT		EXCHANGE	
NewOrderSingle [MsgType 35 = D]1 [TimelnForce 59 = W]	→		
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 0] [ExecType 150 = 0] [TimelnForce 59 = W]	<i>While Connected Order FIX<D> entered with TimelnForce 59=W</i>
Service interruption or MOC disables a participant			
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = I] [ExecType 150 = 4]	<i>A FIX<8> is emitted per While Connected order entered with OrderStatus = 'I' (Eliminated On Disconnect)</i>

While Connected orders cancelled on EOD Mini batch

PARTICIPANT		EXCHANGE	
NewOrderSingle [MsgType 35 = D] [TimelnForce 59 = W]	→		
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 0] [ExecType 150 = 0] [TimelnForce 59 = W]	
EOD Minibatch			

PARTICIPANT		EXCHANGE	
		ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = I] [ExecType 150 = 4]	<i>A FIX<8> is emitted per While Connected order entered with OrderStatus = 'I' (Eliminated On Disconnect)</i>

7.4 Cross/Committed Functionality

Cross/Committed Orders do not interact with the instrument order book. A Cross/Committed Order is submitted:

- NEW ORDER CROSS (35=s) whenever the same member is present on both sides of the order.
- NEW ORDER SINGLE (35=D) and OrdType 40 = C, specifying the opposite firm [Contra Trader 337].

Committed orders traded

PARTICIPANT		EXCHANGE	
NewOrderSingle [MsgType 35 = D] [OrdType 40 = C]	→		
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 0] [ExecType 150 = 0]	
NewOrderSingle [MsgType 35 = D] [OrdType 40 = C]	→		
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 0] [ExecType 150 = 0]	
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 2] [ExecType 150 = 2]	
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 2] [ExecType 150 = 2]	

Committed order cancelled by participant before it trades

PARTICIPANT		EXCHANGE
NewOrderSingle [MsgType 35 = D] [OrdType 40 = C]	→	
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 0] [ExecType 150 = 0]
OrderCancelRequest [MsgType 35 = F]	→	
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 4] [ExecType 150 = 4]

Pending Committed order cancelled during EOD

PARTICIPANT		EXCHANGE
NewOrderSingle [MsgType 35 = D] [OrdType 40 = C]	→	
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 0] [ExecType 150 = 0]
Mini batch, pending one sided orders are flushed		
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 4] [ExecType 150 = 4]

Committed order not accepted by the trading engine

PARTICIPANT		EXCHANGE	
NewOrderSingle [MsgType 35 = D] [OrdType 40 = C]	→		
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 8] [ExecType 150 = 8]	

Entering an Accepted Cross Order

PARTICIPANT		EXCHANGE	
NewOrderCross [MsgType 35 = s]	→		
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 2] [ExecType 150 = 2]	
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 2] [ExecType 150 = 2]	

Cross Order Rejected

PARTICIPANT		EXCHANGE	
NewOrderCross [MsgType 35 = s]	→		
	←	ExecutionReport [MsgType 35 = 8] [OrdStatus 39 = 8] [ExecType 150 = 8]	A NEW ORDER CROSS can be rejected or traded at reception.

7.5 Modification Processing

Modification is accepted

A participant may amend any open orders. Orders that have been fully executed, deleted or cancelled cannot be modified.

A participant cannot modify the instrument ID or the side (buy or sell) of the order. If the quantity is reduced the order retains price/time priority. Any other attributes that are modified will result in the order being eliminated and replaced by a new one.

To modify an order, the participant sends an ORDER CANCEL/REPLACE REQUEST message. In this message, the participant specifies the following elements enabling the Exchange to locate the order:

- The instrument affected by the modification [Symbol 55], [SecurityType 167], [PutOrCall 201], [StrikePrice 202], [MaturityMonthYear 200], [MaturityDay 205] and [OptAttribute 206]
- The [OrigClOrdID 41]
- The side of the order [Side 54]

If the message is valid, the Exchange eliminates the old order from the order book and replaces it with a new one, to which it attributes a new [OrderID 37]. It sends the acknowledgement of the modification in the form of an EXECUTION REPORT message. This message contains the new [OrderID 37] attributed to the modified order by the Exchange and the revised characteristics of the modified order.

The modified order can be:

- Entered in the book (the order has been modified and at least a part of the order has been entered in the Order Book)
- Eliminated (the order has been modified and immediately eliminated)
- Executed partially or in full (the order has been modified and immediately executed in full or partially)

PARTICIPANT		EXCHANGE
Order Cancel/Replace [MsgType 35 = G] [OrdQty 38 = qty]	→	
	←	Execution Report [MsgType 35 = 8] [OrdStatus 39 = 5] [LeavesQty 38 = qty]

Modification is rejected

The Exchange performs validation on the message received. When an error is detected in the incoming message, the Exchange sends a CANCEL REJECT message specifying the error code for the first error detected. If no parameters have been modified, the Exchange sends a CANCEL REJECT specifying 'No modification of the order'.

PARTICIPANT		EXCHANGE	
Order Cancel/Replace [MsgType 35 = G]	→		
	←	Order Cancel Reject [MsgType 35 = 9]	

Cancellation is accepted

Clients may cancel all orders entered by themselves. Cancellations will only be valid for orders, or part of an order, which are currently booked.

To cancel an order, the client sends a CANCEL REQUEST message. This message specifies all the parameters allowing the Exchange to locate the order:

- The instrument affected by the modification [Symbol 55], [SecurityType 167], [PutOrCall 201], [StrikePrice 202], [MaturityMonthYear 200], [MaturityDay 205] and [OptAttribute 206]
- The [OrigClOrdID 41]
- The side of the order [Side 54]

PARTICIPANT		EXCHANGE	
Cancel Request [MsgType 35 = F]	→		
	←	Execution Report [MsgType 35 = 8] [OrdStatus 39 = 4]	<i>If the CANCEL REQUEST is valid, the Exchange will send the acknowledgement of the cancellation in the form of an EXECUTION REPORT ([OrdStatus 39=4]) message specifying the outcome reserved for the order.</i>

Cancellation is rejected

PARTICIPANT		EXCHANGE	
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PARTICIPANT		EXCHANGE	
Cancel Request [MsgType 35 = F]	→		
	←	Order Cancel Reject [MsgType 35 = 9]	Exchange performs validation on the message received. If the CANCEL REQUEST is not valid, the Exchange sends a CANCEL REJECT message indicating the error code for the first error detected.

7.6 Strategy Messages

User defined Strategy (FLEXCO) Creation Accepted

A trader can request the creation of a user defined strategy (UDS) by submitting a SECURITY DEFINITION REQUEST. The SECURITY DEFINITION REQUEST must include the strategy leg information.

Strategy Creation requests can be accepted [SecurityResponseType 321=1], accepted but modified [SecurityResponseType 321=2] or rejected [SecurityResponseType 321=3].

If the creation is modified the SECURITY DEFINITION message includes the new instrument structure. The leg ordering sequence may differ from the original request but will not be marked as modified if the ratio and the verb for all legs remain the same.

PARTICIPANT		EXCHANGE	
Security Definition Request [MsgType 35 = c]	→		
	←	Security Definition [MsgType 35 = d]	

User defined Strategy (FLEXCO) Creation Error

PARTICIPANT		EXCHANGE	
Security Definition Request [MsgType 35 = c]	→		
	←	Error [MsgType 35 = 3]	

7.7 Quote Processing

Request for Quote

Request for Quote entry allows participants to broadcast a REQUEST FOR QUOTE to other participants.

The client enters a QUOTE REQUEST message with the symbol and an optional quantity. If the message is valid, the participant receives a QUOTE ACKNOWLEDGEMENT message with [QuoteAckStatus 297=0]. If the QUOTE REQUEST message is not valid, the Exchange sends a QUOTE ACKNOWLEDGEMENT message with [QuoteAckStatus 297=5].

PARTICIPANT		EXCHANGE
Quote Request [MsgType 35 = R]	→	
	←	Quote Acknowledgement [MsgType 35 = b]

7.8 Unsolicited Services

Entry or Cancellation of an Order by the Exchange.

The Exchange may enter or cancel orders on behalf of a participant.

The cancellation can be done for orders entered by the Exchange in the participant's account. This action can take place during:

- Order Cancellation
- Trading Session

If the Exchange enters an order on behalf of a participant, the client does not receive any acknowledgements or receive any messages related to this order.

The participant that entered the initial order will receive the EXECUTION REPORT [OrdStatus 39=4 or 5] message for any orders cancelled by the Exchange.

Elimination of an Order

The table below describes all the scenarios where order elimination may occur without the participant sending a cancellation message.

Reason for Elimination	
The order price is outside the instrument limit price	EXECUTION REPORT [OrdStatus 39=F or G] message sent to the Participant who initially entered the order

Reason for Elimination	
During the instrument opening Market Order without opposite order are eliminated.	EXECUTION REPORT [OrdStatus 39=4] message sent to the Participant who initially entered the order
Participant disconnection eliminates While Connected Order.	EXECUTION REPORT [OrdStatus 39=I] message sent to the Participant who initially entered the order
Instrument state does not allow order with disclosed quantity.	EXECUTION REPORT [OrdStatus 39=4] message sent to the Participant who initially entered the order
Cancellation of an order by the Exchange	Possible during: Order Cancellation, Trading Session, Exchange Intervention, Consultation End of Day, and Group Interruption. EXECUTION REPORT [OrdStatus 39=4] message sent to the Participant who initially entered the order
Instrument has expired, updated or is deleted	Carried out during Mini Batch or Post-session. If the TICK INCREMENT for prices is modified, all prices that do not respect the new TICK INCREMENT are purged. The others remain on the order book.
Validity of the order is reached	Carried out at the end of each trading day just before or during Maintenance. Also carried out at the end of the week (last trading day of the weekly session) just before Post-session
Self Execution Prevention is triggered	EXECUTION REPORT [OrdStatus 39=5] message is sent to the Participant whose order has been eliminated due to Self Execution Prevention (SEP). Order Updates are notified with TAG 84 (CxlQty) in the EXECUTION REPORT message indicating the quantity removed by Self Execution Prevention (SEP).

Global Cancellation of all Orders for a Member Initiated by the Exchange

PARTICIPANT		EXCHANGE	
	←	Execution Report [MsgType 35=8] [OrdStatus 39=4]	<i>Will be sent for every booked order posted by the member</i> <i>No [OrigClOrdID 41] is sent</i>

Cancellation of a Trade by the Exchange

If required, the Exchange can cancel a trade that took place during the day. This cancellation can be initiated in accordance with the Exchange rules.

This can take place during:

- Order Cancellation
- Pre-Opening
- Trading Session

- Suspended
- Exchange Intervention
- Consultation End
- On an Interrupted group

The Exchange sends the two clients concerned an EXECUTION REPORT ([ExecTransType 20=1]) message. This message specifies all the parameters relating to the cancelled trade.

If the trade involved a strategy instrument, the Exchange will cancel a trade on a leg-by-leg basis. For each leg trade cancellation, the Exchange sends an EXECUTION REPORT ([ExecTransType 20=1, MultiLegReportingType=2]) to each of the two Participants. When all the legs-trades have been cancelled, the Exchange also sends an EXECUTION REPORT ([ExecTransType 20=1, MultiLegReportingType=3]) to the participant(s) who entered the strategy order.

Cancellation of a Trade with Impact on the Last Price Made by the Exchange

PARTICIPANT		EXCHANGE
	←	Execution Report [MsgType 35=8] [OrdStatus 39=4] [ExecTransType 20=1]

7.9 Queries

In order to assist clients manage and synchronize their trading database, the Exchange offers two types of queries: Order Mass Status and Security Definition.

Order Mass Status

This query returns all orders that were entered during the day for the participant. One EXECUTION REPORT [ExecTransType 20=3] is returned for each active order of the Participant. The maximum number of requests is limited to 5 per Participant per day.

Security Definition

This query returns all listed instruments on the Exchange. This request generates one message per instrument. For strategies, SECURITY DEFINITION responses include the definition of each leg.

The maximum number of requests is limited to 3 per participant per day and they are only permitted before the Market Opening.